Step - 1 : Import Libraries

In [4]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

Step-2: Load Dataset

In [5]:
```
df = pd.read_csv("../data/DisasterDeclarations.csv")

df.head()
```

Out[5]:

	femaDeclarationString	disasterNumber	state	declarationType	declarationDate	fyDecla
0	FM-5529-OR	5529	OR	FM	2024-08-09T00:00:00.000Z	2
1	FM-5528-OR	5528	OR	FM	2024-08-06T00:00:00.000Z	2
2	FM-5527-OR	5527	OR	FM	2024-08-02T00:00:00.000Z	2
3	DR-4312-CA	4312	CA	DR	2017-05-02T00:00:00.000Z	2
4	DR-4251-AL	4251	AL	DR	2016-01-21T00:00:00.000Z	2

5 rows × 28 columns

Step - 3:Validate Data

In [6]:
```
df['state'].unique()
df['state'].isnull().sum()
```

Out[6]:
```
np.int64(0)
```

Step - 4: State-Level Aggregation

In [7]:
```
state_counts = df['state'].value_counts()

state_counts.head()
```

Out[7]:
```
state
TX    5388
KY    3355
MO    2830
FL    2791
GA    2765
Name: count, dtype: int64
```

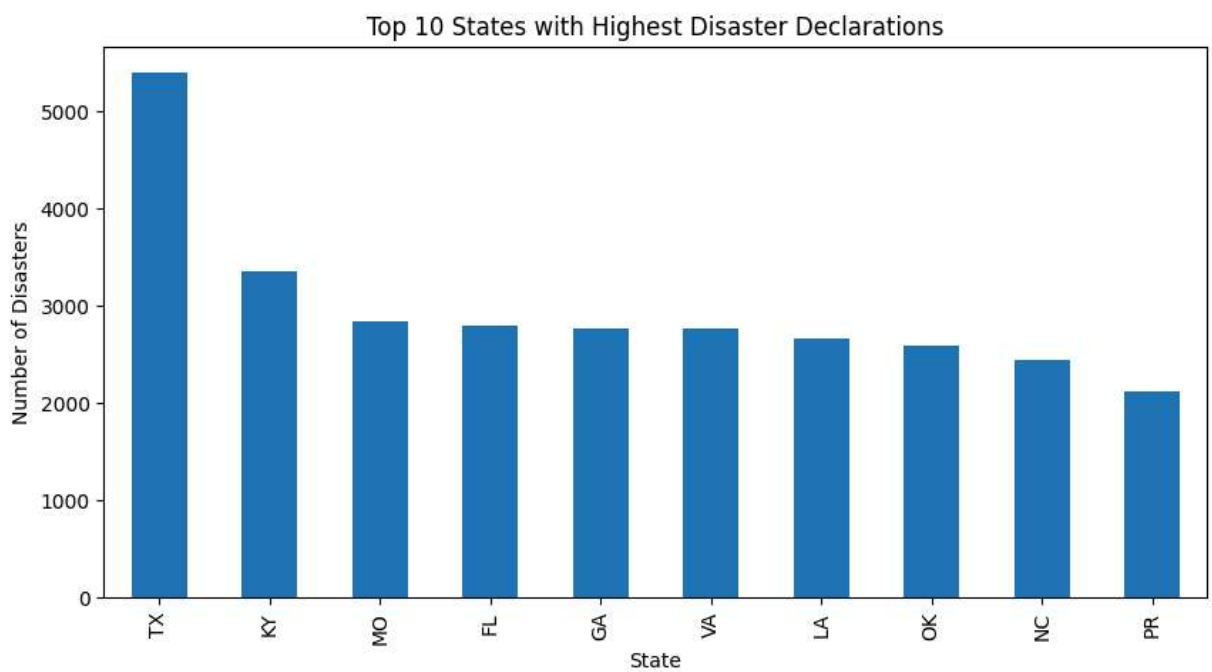
Step -5: Visualization 1: Top 10 Disaster-Prone States

In [8]:
```
plt.figure(figsize=(10,5))

state_counts.head(10).plot(kind='bar')

plt.title("Top 10 States with Highest Disaster Declarations")
plt.xlabel("State")
plt.ylabel("Number of Disasters")

plt.show()
```



Insight:

Observation: The chart shows the states with the highest number of disaster declarations.

Pattern: States like Texas, California, and Florida have the highest counts.

Reason: These states are more exposed to hurricanes, wildfires, and floods.

Step - :6 Choropleth Map

In [9]:
```
state_df = state_counts.reset_index()
state_df.columns = ['state', 'count']

fig = px.choropleth(
    state_df,
    locations='state',
    locationmode="USA-states",
    color='count',
    scope="usa",
    color_continuous_scale="Reds",
    title="Disaster Declarations by State"
)

fig.show()
```

Insight:

Observation: The choropleth map shows disaster distribution across US states.

Pattern: Southern and coastal states show higher disaster density.

Reason: These regions are more vulnerable to hurricanes, floods, and storms.

Step - 7: State + Incident Type Analysis

In [12]:
```
state_incident = df.groupby(['state','incidentType']).size().unstack().fillna(0)

state_incident.head()
```

Out[12]:

incidentType	Biological	Chemical	Coastal Storm	Dam/Levee Break	Drought	Earthquake	Fire	Fishin Losse
state								
AK	121.0	0.0	2.0	0.0	0.0	13.0	35.0	0.
AL	139.0	0.0	0.0	0.0	67.0	0.0	11.0	0.
AR	152.0	0.0	0.0	0.0	32.0	0.0	0.0	0.
AS	10.0	0.0	0.0	0.0	5.0	5.0	0.0	0.
AZ	73.0	0.0	0.0	0.0	8.0	0.0	97.0	0.

5 rows × 27 columns

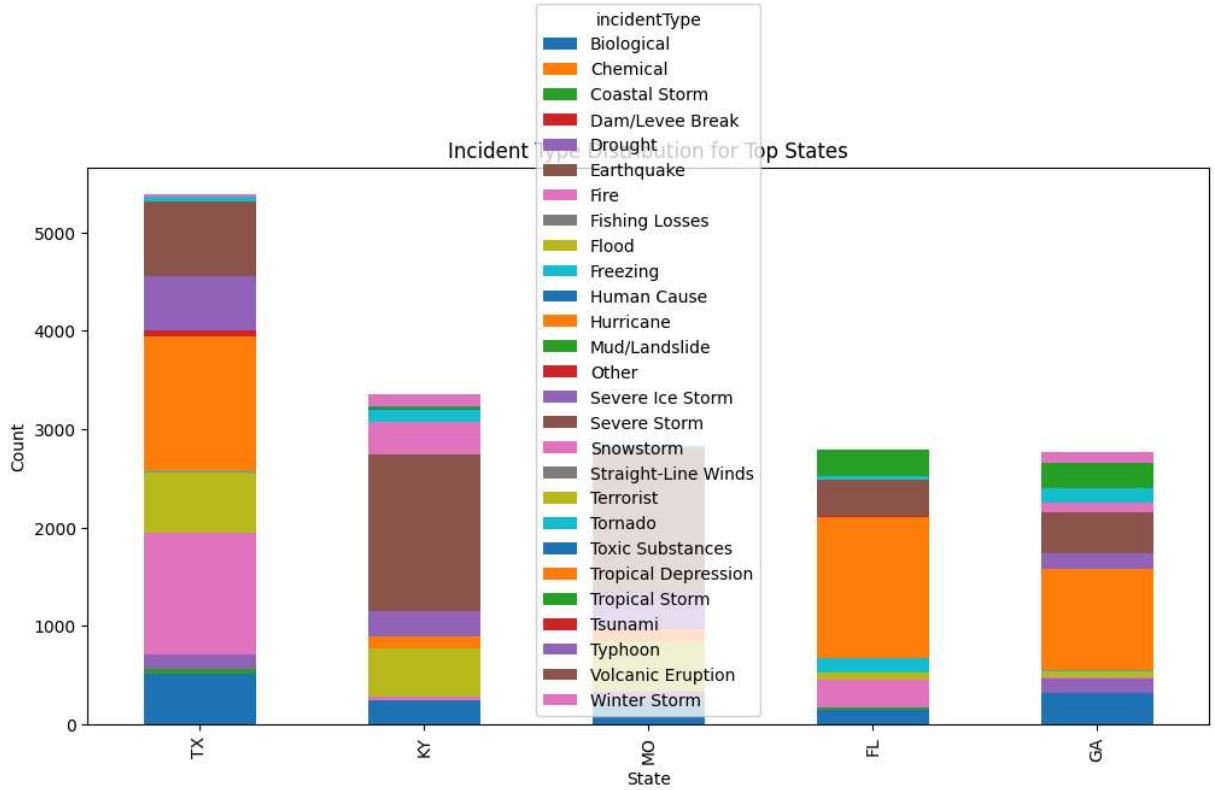
Step - 7: Stacked Bar Chart

In [13]:
```
state_incident.loc[state_counts.head(5).index].plot(
    kind='bar',
    stacked=True,
```

```
    figsize=(12,6)
)

plt.title("Incident Type Distribution for Top States")
plt.xlabel("State")
plt.ylabel("Count")

plt.show()
```



Insight:

Observation: The stacked chart shows how different disaster types contribute per state.

Pattern: Storms dominate most states, while fires are more common in California.

Reason: Geographical conditions influence disaster types in each region.

Step - 8: Hotspot Map for Specific Disaster

In [14]:
```
flood_df = df[df['incidentType'] == 'Flood']

flood_counts = flood_df['state'].value_counts().reset_index()
flood_counts.columns = ['state', 'count']

fig = px.choropleth(
    flood_counts,
    locations='state',
    locationmode="USA-states",
    color='count',
    scope="usa",
```

```
    color_continuous_scale="Blues",
    title="Flood Disaster Hotspots"
)

fig.show()
```